Simulation Interoperability Standards Organization

"Simulation Interoperability & Reuse through Standards"

Ensuring Time Resolution Compatibility for Simulation Executives, Job Scheduling, and Data Exchange

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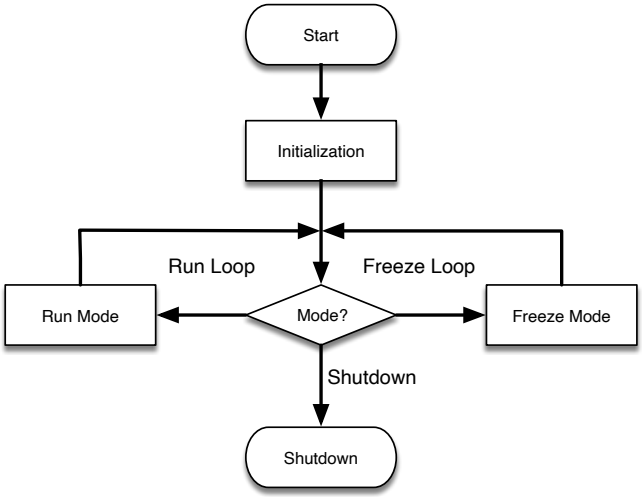
Introduction

- **This paper is a prequel to our 2025 SIW Paper [1]: *A Discussion of Time Management Concepts and Time Constraint Equations for Multi-Rate Federation Executions***
- **Managing and modeling time in digital simulations is often more complicated than most people think.**
- **We discuss some key concepts associated with time and time resolution in digital time-based physics-based simulations.**
- **We discuss some of the consequences of not have compatible time resolutions.**
- **We derive specific constraints that can be tested to ensure time resolution compatibility in:**
  - Simulation executives,
  - Job scheduling, and
  - Data exchange.
- **We look at some specific constraints for HLA-based distributed simulations.**
- **Additional real-time considerations are also discussed in the paper but not presented here, due to time constraints.** 😀

Concepts and Definitions: Execution



- ***Jobs***: Specific model functions called in the execution loops.
- ***Frames***: Specified cycles in the execution loops that collections of jobs are scheduled to run.
- ***Initialization***: The entry mode for a simulation where methods are called to initialize the state of the simulation executive and all the constituent models. Timelines are typically established and defined here but time is not advancing.
- ***Run***: Generally, the principal execution loop in a simulation where model jobs are called and timelines are coordinated and advanced.
- ***Freeze***: An alternate execution loop in a simulation where special "freeze" jobs are called. While many of the simulation related timelines are not advanced in this mode, the simulation executive may maintain and manage a freeze loop executive counter.
- ***Shutdown***: The exit mode for a simulation where methods are called to close out the simulation executive, safe any related systems, release resources, and terminate the program.

Concepts and Definitions: Timelines

- **There are 4 principal timelines associated with most time-based digital simulations:**
  - **Physical Time (PT):** the non-spatial dimension associated with our spacetime continuum in which events are ordered in irreversible succession from the past to the present to the future.
  - **Computer Clock Time (CCT):** the representation of time maintained and managed by the computer usually using some form of oscillator counting oscillations from a known epoch and mapped into an approximation of PT.
  - **Simulation Elapsed Time (SET):** the time measure associated with an individual simulation starting at zero and advancing monotonically in quantifiable steps.
  - **Simulation Scenario Time (SST):** a model within a simulation that associates the Simulation Elapsed Time with a representation of the problem's Physical Time.
- **There are 2 additional timelines associated with HLA-based distributed simulations:**
  - **HLA Logical Time (HLT):** the timeline used by HLA to order messages, regulate execution time advance (Time Advance Request (TAR) & Time Advance Grant (TAG)) and enable deterministic behavior in distributed simulation.
  - **Federation Scenario Time (FST):** a time associated with the physical systems being modeled in the participating federates in the federation execution.

Concepts and Definitions: Time Resolution

- **Note the following definitions related to executive time and Ticks:**
  - t_{se} is the current Simulation Elapsed Time (SET);
  - τ represents a single executive time Tick (discrete unit of measure);
  - $n_\tau \equiv T$ is the number of executive Ticks since the start of the simulation;
  - M_e is the fractional relationship between executive time Ticks and seconds (a.k.a., Tick rate or Tick Multiplier); and
  - $\epsilon \equiv dt_\tau$ is time step associated with a single executive time Tick.
- **The executive time to Tick relationship is: $t_{se} = T/M_e$ or $T = t_{se}\, M_e$**
- **Note that the minimum time resolution is: $\epsilon \equiv dt_\tau \equiv 1/Me$**
  - The executive cannot represent times any smaller than this.

Time Representations: Background

- **Refer to the excellent 2011 SIW paper [8]: *Time Representation and Interpretation in Simulation Interoperability - An Overview***
  - Provides an excellent overview of time representations.
  - Key Take Aways:
    - *Understand the limitations of the time representation that you intend to use.*
    - *Use time representations that have a big enough value range for your simulation.*
    - *Consider using integer or fixed-point time representations.*
    - *Use the standardized time types according to HLA Evolved (IEEE 1516-2010).*
- **We will be discussing integer-based simulation executive times.**

Time Representations: Job Cycle Times and Tick Multipliers

- **In general, a job cycle time can be expressed as: $dt_\omega = N_\omega / D_\omega$**
  - Where dt_ω is the job cycle time expressible as a rational fraction,
  - N_ω and D_ω are the integer numerator and denominator composing the fraction, and
  - ω is the corresponding job cycle frequency.
- **A minimum executive time resolution (Tick) can be derived that will ensure compatibility between executive time advance and job/frame scheduling:**
  - Determine the job cycle time ratios for all jobs.
  - Find the Least Common Denominator for all job cycle times using prime factorization: D_p.
  - This will be the minimum executive time Tick Multiplier: $M_p = D_P$.
- **A compatible Tick Multiplier (M_e) can be any integer multiple of M_p.**

Time Representations: Compatibility Example

- **Let's look at an example.**
- **Our simulation has the following job cycle frequencies (Hz): 10, 30, 100, and 128.**
  - Results in cycle time ratios (dt_ω): 1/10, 1/30, 1/100, and 1/128.
  - Prime factorization gives: 960/9600, 320/9600, 96/9600, and 75/9600.
- **The minimum time Tick Multiplier: $M_p = 9600$**
  - The resulting time resolution is: $\epsilon \cong 0.00010417$
  - Probably not sufficient for most applications.
  - Fortunately, the executive Tick Multiplier can be any integer multiple of the minimum Tick Multiplier: $M_e = N\ M_p$
- **Assume we want a time resolution better than 1 µs.**
  - $M_e = 200\ M_p = 1{,}920{,}000$
  - $\epsilon \cong 0.00000052$
  - Better than 1 µs
- **An $M_e = 1{,}920{,}000$ will be compatible with all job cycle times and meet the executive time resolution requirements.**

Consequences

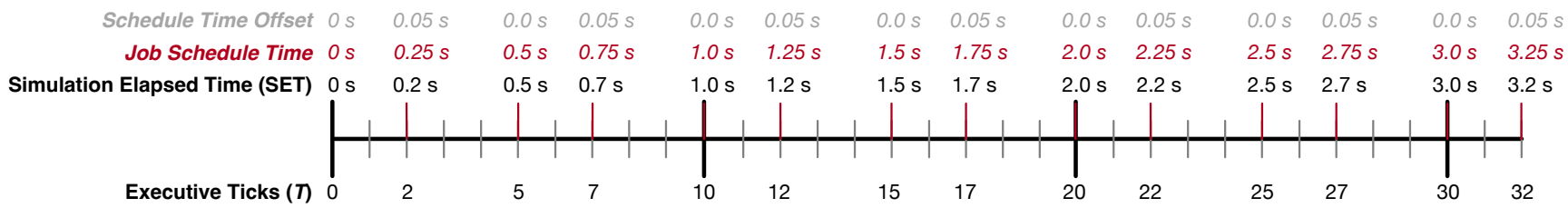
- **What happens if the executive time resolution and the job cycle times are not compatible?**
  - In general, **incompatibility** means the executive time Tick computation does not result in an integer number of executive time Ticks.
  - Executives can handle this in many ways.
- **We can look at 2 variations of 2 approaches:**
  - Simulation Elapsed Time (SET): $T = t_{se} * M_e$
    - $T = \textbf{trunc}(t_{se} * M_e)$
    - $T = \textbf{round}(t_{se} * M_e)$
  - Time step accumulation: $T = (n_\tau * dt_\omega) * M_e$
    - n_τ is the integer number of Ticks since the beginning of the simulation.
    - $T = n_\tau * \textbf{trunc}(dt_\omega * M_e)$
    - $T = n_\tau * \textbf{round}(dt_\omega * M_e)$

Consequences: Example

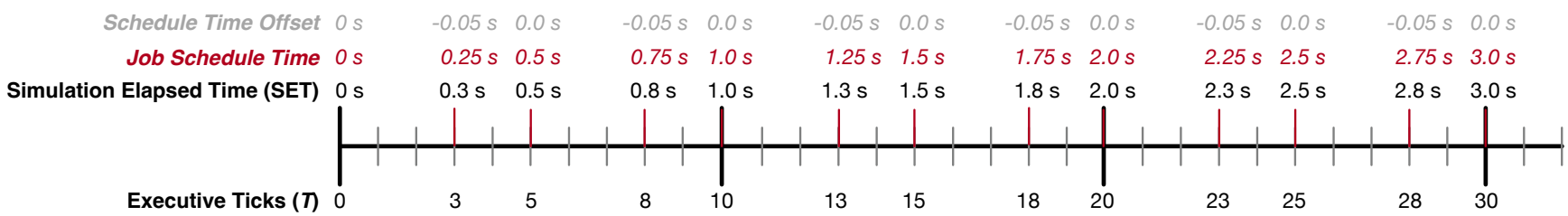
- **Again, a simple example will help.**
  - Simulation executive Tick multiplier: $M_e = 10$ ($\epsilon = 0.1$).
  - A single job scheduled at 4 Hz: $dt_\omega = 1/4 = 0.25$
  - M_e and dt_ω are not compatible.
- **Let's look at the four executive time computation approaches:**
  - Simulation Elapsed Time (SET): $T = t_{se} * M_e$
    - $T = \textbf{trunc}(t_{se} * M_e)$
    - $T = \textbf{round}(t_{se} * M_e)$
  - Time step accumulation: $T = (n_\tau * dt_\omega) * M_e$
    - $T = n_\tau * \textbf{trunc}(dt_\omega * M_e) = n_\tau * 0.2$
    - $T = n_\tau * \textbf{round}(dt_\omega * M_e) = n_\tau * 0.3$

Consequences: SET

- **Simulation Elapsed Time (SET): $T = t_{se} * M_e$**
  - $T = \textbf{trunc}(t_{se} * M_e)$: Times match sometimes and not others. Early when not matched.



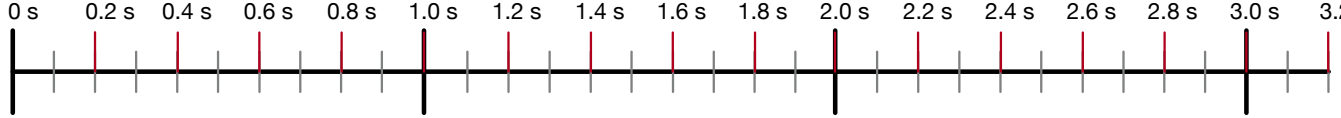
  - $T = \textbf{round}(t_{se} * M_e)$: Times match sometimes and not others. Late when not matched.



Consequences: Accumulation

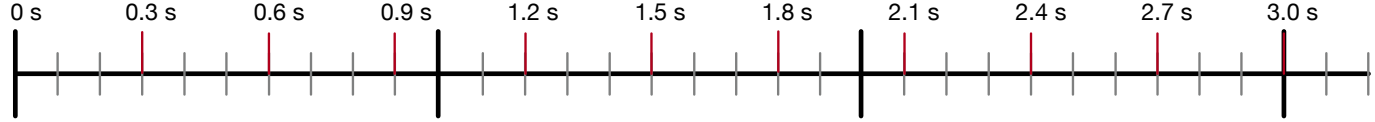
- **Time step accumulation: $T = (n_\tau * dt_\omega) * M_e$**
  - $T = n_\tau * \textbf{trunc}(dt_\omega * M_e) = n_\tau * 0.2$: Increasingly runs ahead.

Schedule Time Offset	0 s	0.05 s	0.1 s	0.15 s	0.2 s	0.25 s	0.3 s	0.35 s	0.4 s	0.45 s	0.5 s	0.55 s	0.6 s	0.65 s	0.7 s	0.75 s	0.8 s
Job Schedule Time	*0 s*	*0.25 s*	*0.5 s*	*0.75 s*	*1.0 s*	*1.25 s*	*1.5 s*	*1.75 s*	*2.0 s*	*2.25 s*	*2.5 s*	*2.75 s*	*3.0 s*	*3.25 s*	*3.5 s*	*3.75 s*	*4.0 s*
Simulation Elapsed Time (SET)	0 s	0.2 s	0.4 s	0.6 s	0.8 s	1.0 s	1.2 s	1.4 s	1.6 s	1.8 s	2.0 s	2.2 s	2.4 s	2.6 s	2.8 s	3.0 s	3.2 s
Executive Ticks (*T*)	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32



  - $T = n_\tau * \textbf{round}(dt_\omega * M_e) = n_\tau * 0.3$: Increasingly lags behind.

Schedule Time Offset	0 s	-0.05 s	-0.1 s	-0.15 s	-0.2 s	-0.25 s	-0.3 s	-0.35 s	-0.4 s	-0.45 s	-0.5 s
Job Schedule Time	*0 s*	*0.25 s*	*0.5 s*	*0.75 s*	*1.0 s*	*1.25 s*	*1.5 s*	*1.75 s*	*2.0 s*	*2.25 s*	*2.5 s*
Simulation Elapsed Time (SET)	0 s	0.3 s	0.6 s	0.9 s	1.2 s	1.5 s	1.8 s	2.1 s	2.4 s	2.7 s	3.0 s
Executive Ticks (*T*)	0	3	6	9	12	15	18	21	24	27	30



Distributed Simulation

- **How are distributed simulation and simulation data exchange impacted by time resolution? What constitutes compatibility?**
- **HLA Considerations:**
  - Each HLA Federate is a simulation with its own time Tick (τ_f).
  - Therefore, each HLA Federate has a Tick Multiplier (M_f).
  - The Federation Execution as a whole has a time Tick (τ_{hlt}) derived from the base time resolution for the HLA Logical Time (HLT).
  - This implies an HLA Federation Execution Tick Multiplier (M_{hlt}).
- **Federate/Federation Multiplier Compatibility Requirements:**
  - $M_f \geq M_{hlt}$: The federate must have sufficient time resolution.
  - $M_f \% M_{hlt} = 0$: The federate multiplier must be an integer multiple.

Distributed Simulation: Event Timing

- **There is an additional consideration for HLA event timing.**
- **Events will be mapped to federate and federation time:**
  - $T_{hlt} = (T_f * M_{hlt})/M_f$
  - $T_f = (T_{hlt} * M_f)/M_{hlt}$
- **Note that these time relationships are directly related to the ratio of the federate to federation Tick Multipliers: $R_f \equiv M_f / M_{hlt}$**
  - R_f will be an integer since $M_f \geq M_{hlt}$ and $M_f \% M_{hlt} = 0$.
  - $T_f = T_{hlt} * R_f$
- **How does this restrict compatible event times?**
  - $T_{f_event} \% R_f = 0$
  - Specifically, federate generated event times are limited to integer multiples of the ratio of the federate to federation Tick Multipliers (R_f).

Conclusions

- **Executive Tick Multiplier (M_e) should be compatible with job cycle times (dt_ω).**
- **Use Prime Factorization to compute a compatible base Tick Multiplier: M_p.**
- **Higher resolution compatible M_e will be any integer multiple of M_p.**
- **HLA-based distributed simulations also have a time resolution: M_{hlt}.**
- **HLA-based federate time resolutions must be compatible with the federation time resolution:**
  - $M_f \geq M_{hlt}$
  - $M_f \% M_{hlt} = 0$
- **HLA federate time events are limited to integer multiples of the ratio of the federate to federation Tick Multiplier: $T_{f_event} \% R_f = 0$.**

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Q&A / Discussion

Key References

[1] E. Crues and D. Dexter. A Discussion of Time Management Concepts and Time Constraint Equations for Multi-Rate Federation Executions. In SISO 2025 Simulation Innovation Workshop. SISO, SISO, February 2025.

[2] Simulation Interoperability Standards Organization/ Standards Activities Committee (SISO/SAC). IEEE Standard for Modeling and Simulation (M&S) High Level Architecture (HLA) - Framework and Rules. Technical Report IEEE-1516-2010, The Institute of Electrical and Electronics Engineers, 2 Park Avenue, New York, NY 10016-5997, August 2010.

[5] E. Crues, D. Dexter, A. Falcone, A. Garro, M. Madden, B. Möller, and D. Ronnfeldt. Standard for Space Reference Federation Object Model (SpaceFOM). Number SISO-STD-018-2020. Simulation Interoperability Standards Organization (SISO), P.O. Box 781238, Orlando, FL 32878-1238, USA, October 2019.

[7] National Aeronautics and Space Administration, Johnson Space Center, Software, Robotics & Simulation Division, Simulation and Graphics Branch (ER7). Trick simulation environment: Documentation. Trick Wiki Site: https://nasa.github.io/trick/documentation/Documentation-Home, September 2023.

[8] M. Karlsson, F. Antelius, and B. Möller. Time Representation and Interpretation in Simulation Interoperability - An Overview. In SISO 2011 Spring Simulation Interoperability Workshop. SISO, SISO, April 2011.